Supplemental information

HistoAtlas: an open, interpretable pan-cancer histomics resource linking tissue morphology to molecular programs and clinical outcomes

Pierre-Antoine Bannier

This document (Document S1) contains Figures S1–S4 and Tables S1–S13. Tables marked *machine-readable* are released in full with the atlas rather than typeset here, because they contain between 10^3 and 10^5 rows; the file path given for each is relative to the root of the code repository, and every one of them is also queryable through the web and programmatic interfaces described in the main text. All statistics were produced by the pipeline described in the main-text Methods; nothing in this document was recomputed for presentation.

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Supplementary Figures

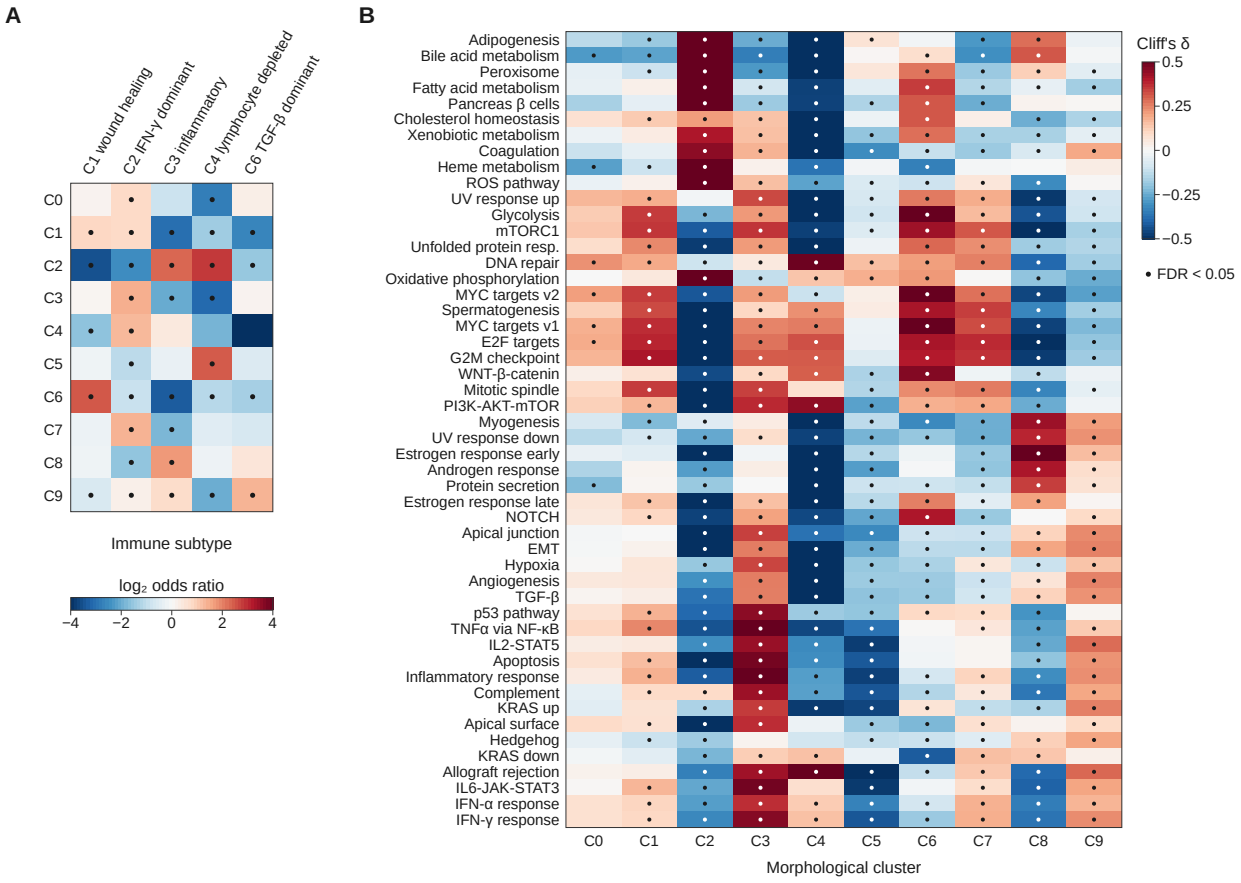


Figure S1: Morphological clusters are enriched for distinct immune and pathway states. (A) Immune subtype enrichment per pan-cancer morphological cluster: $\log_2$ odds ratio of each Thorsson immune subtype within the cluster versus the rest of the cohort (Fisher's exact test), clipped at ± 4. Dots mark Benjamini–Hochberg adjusted $P < 0.05$. (B) Hallmark pathway enrichment per cluster (Cliff's δ from Mann–Whitney U tests, in-cluster versus out-of-cluster). Pathways are ordered by Ward-linkage clustering; dots mark Benjamini–Hochberg adjusted $P < 0.05$. These enrichments were computed after clustering and were not used to define the clusters, which were derived from the 38 histomic measurements alone.

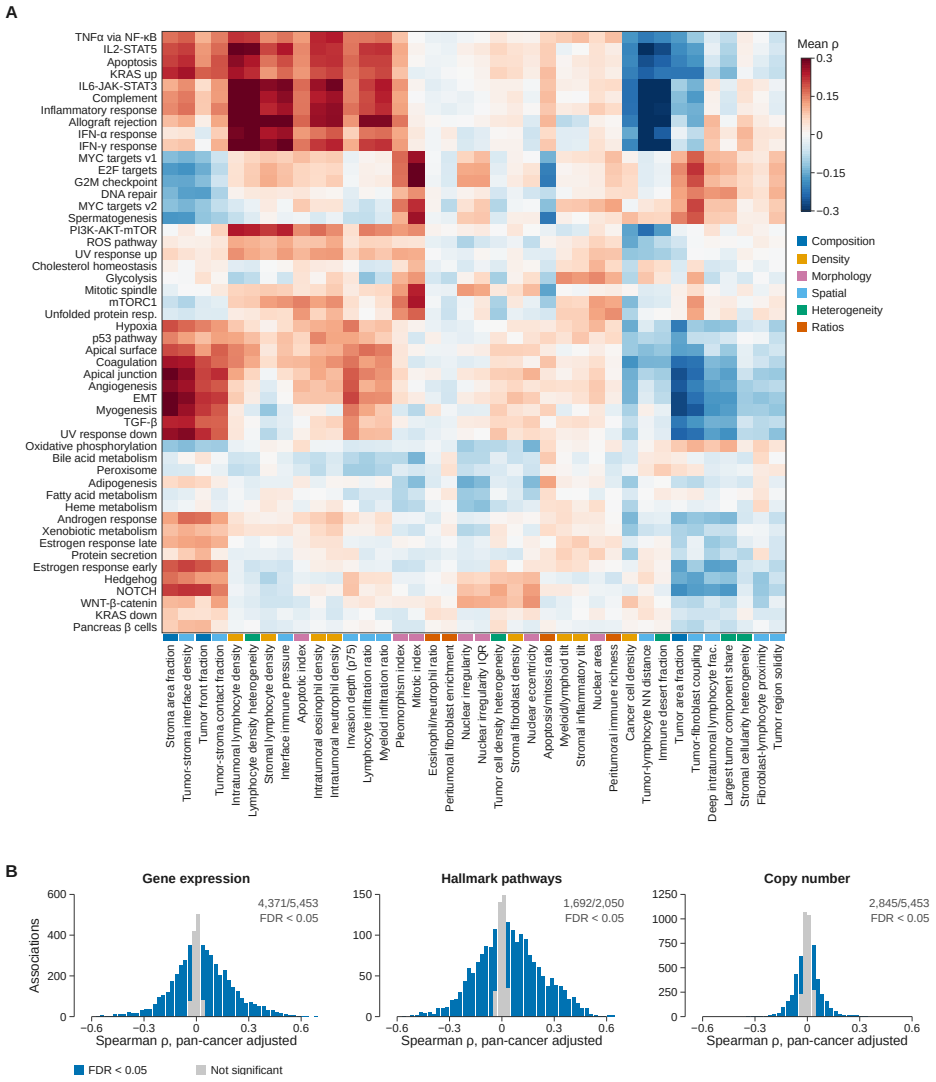


Figure S2: Histomic measurements correspond systematically to Hallmark pathway activity. (A) Mean Spearman correlation across the 21 cancer types between each of the 38 histomic measurements (columns) and each of the 50 MSigDB Hallmark pathway activity scores (rows), unadjusted model. Rows and columns are ordered by Ward-linkage clustering; the strip below the columns gives the measurement group. Immune-density measurements group with immune and inflammatory signatures; nuclear morphometry and mitotic index with cell-cycle and proliferation signatures; stromal and tumor–stroma interface measurements with epithelial–mesenchymal transition, angiogenesis and coagulation signatures. Color scale clipped at $\rho = \pm 0.3$. (B) Pan-cancer covariate-adjusted Spearman coefficients by molecular data type, stacked histograms of significant (blue, false discovery rate below 0.05) and non-significant (gray) associations: gene expression, 4,371 of 5,453 significant (80%); Hallmark pathways, 1,692 of 2,050 (83%); copy number, 2,845 of 5,453 (52%). P values from the analytical t-approximation, Benjamini–Hochberg corrected within declared families.

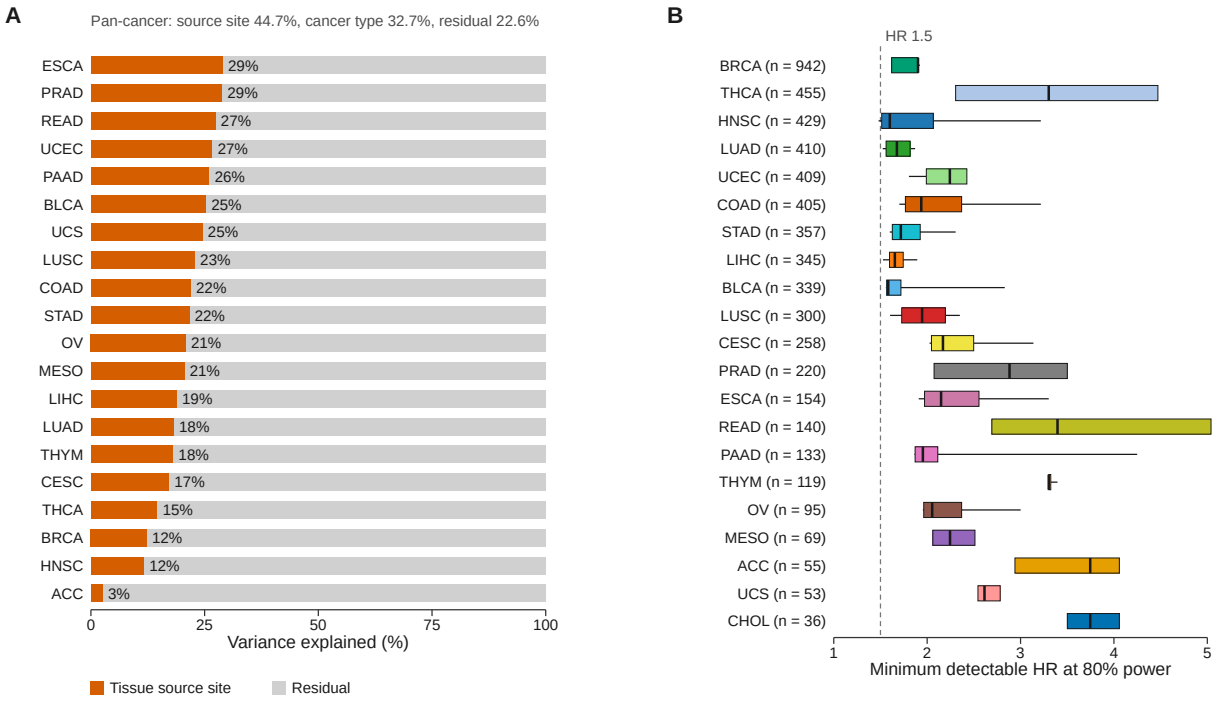


Figure S3: Batch effects and statistical power. (A) Principal variance component analysis per cancer type: proportion of variance in the histomic measurements attributable to tissue source site (orange) and residual variance (gray), for the 20 cancer types with more than one source site. At pan-cancer level, 44.7% of variance is attributed to source site, 32.7% to cancer type and 22.6% to residual; within cancer types, source site accounts for a median of 21.3%. All per-cancer silhouette scores computed on source-site labels are negative (-0.18 to -0.0004), so slides do not segregate by source site in measurement space. (B) Minimum detectable hazard ratio at 80% power ($\alpha = 0.05$, two-sided) for harmful effects, by cancer type ordered by sample size (n, median patients per model). Boxes show the interquartile range across survival models, the bar the median and whiskers the 5th to 95th percentiles; dashed line, HR = 1.5. Breast carcinoma can detect HR ≥ 1.62; cholangiocarcinoma ($n = 36$) requires HR ≥ 3.75. Power, not absence of effect, explains why 4,846 of 5,623 survival associations carry an *insufficient* evidence badge.

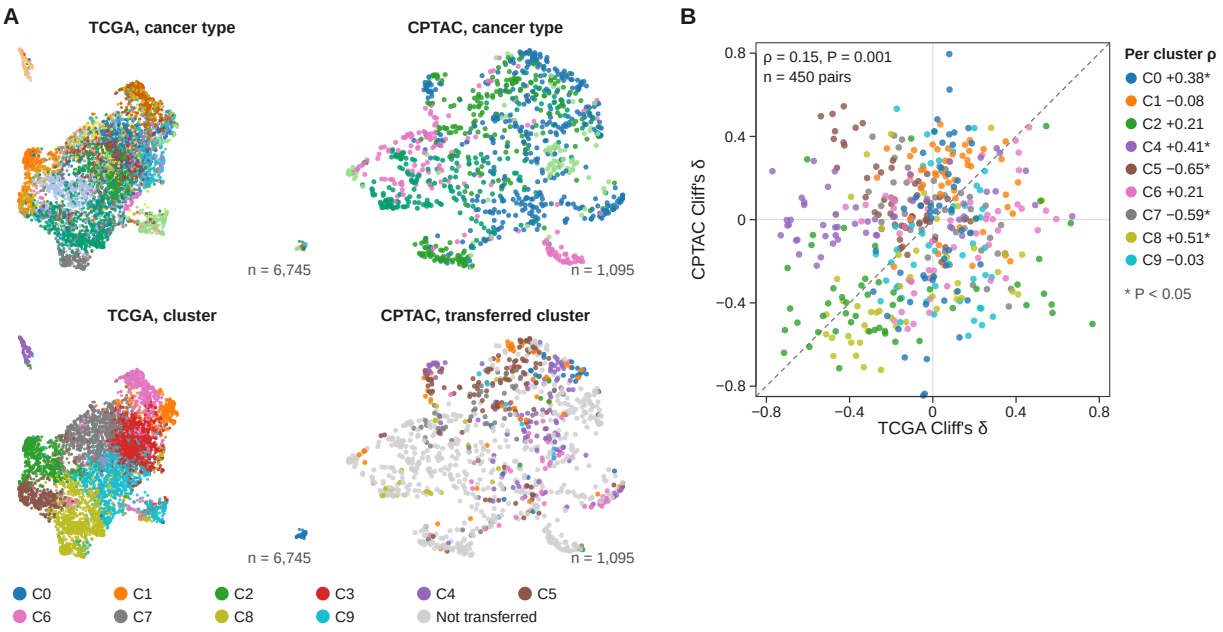


Figure S4: Hard cluster labels transfer only partially to an independent cohort. (A) UMAP embeddings of TCGA (n = 6,745 slides, left) and CPTAC (n = 1,095 slides, right; a separate embedding computed on CPTAC), colored by cancer type (top; colors as in Figure 1C, HNSCC as HNSC) and by pan-cancer morphological cluster (bottom). CPTAC cluster labels were assigned by nearest-centroid prediction from the K-means model fitted on TCGA without refitting; slides without a complete measurement vector received no label (gray). (B) Per-cluster pathway enrichment (Cliff's δ) computed independently in TCGA (horizontal axis) and CPTAC (vertical axis) for 450 matched cluster–pathway pairs (no CPTAC slide was assigned to C3); dashed line, identity. Overall Spearman $\rho = 0.15$, $P = 1.4 \times 10^{-3}$. The list gives the per-cluster Spearman coefficient over 50 pathways; asterisks mark $P < 0.05$. Three clusters transfer positively ($\rho = 0.38$ to 0.51) and two invert ($\rho = -0.59$ and -0.65). This result is why the main text treats the named measurements and the continuous programs, rather than the discrete clusters, as the portable products of the atlas.

Supplementary Tables

Table S1: Composition of the TCGA discovery cohort. Diagnostic whole-slide images after quality control, and the patients they come from. All slides enter the descriptive analyses; statistical associations use one representative slide per patient.

Code	Cancer type	Slides	Patients
BRCA	Breast invasive carcinoma	1,037	973
LUAD	Lung adenocarcinoma	511	448
THCA	Thyroid carcinoma	473	462
HNSC	Head and neck squamous cell carcinoma	471	449
UCEC	Uterine corpus endometrial carcinoma	459	421
COAD	Colon adenocarcinoma	441	433
BLCA	Bladder urothelial carcinoma	417	350
STAD	Stomach adenocarcinoma	400	375
LIHC	Liver hepatocellular carcinoma	365	357
LUSC	Lung squamous cell carcinoma	357	340
PRAD	Prostate adenocarcinoma	353	319
CESC	Cervical squamous cell carcinoma	279	269
ACC	Adrenocortical carcinoma	227	56
THYM	Thymoma	180	121
ESCA	Esophageal carcinoma	158	156
READ	Rectum adenocarcinoma	157	156
PAAD	Pancreatic adenocarcinoma	146	135
OV	Ovarian serous cystadenocarcinoma	107	106
UCS	Uterine carcinosarcoma	87	53
MESO	Mesothelioma	82	71
CHOL	Cholangiocarcinoma	38	38
Total (21 cancer types)		6,745	6,088

Table S2: TCGA cancer types excluded from the atlas. All twelve were excluded because their dominant cell morphologies fall outside the training domain of the cell detection model, which was developed on epithelial and squamous tissue. Exclusion was decided from the model's documented training composition, before any downstream analysis.

Code	Cancer type	Dominant morphology outside training domain
DLBC	Diffuse large B-cell lymphoma	Lymphoid cells
LAML	Acute myeloid leukemia	Myeloid blasts
GBM	Glioblastoma multiforme	Glial cells
LGG	Brain lower-grade glioma	Glial cells
SKCM	Skin cutaneous melanoma	Melanocytes
UVM	Uveal melanoma	Melanocytes
SARC	Sarcoma	Mesenchymal cells
PCPG	Pheochromocytoma and paraganglioma	Neuroendocrine cells
KIRC	Kidney renal clear cell carcinoma	Renal tubular cells
KIRP	Kidney renal papillary cell carcinoma	Renal tubular cells
KICH	Kidney chromophobe	Renal tubular cells
TGCT	Testicular germ cell tumors	Germ cells

Twenty-one of the 33 TCGA projects are included (Table S1) and these twelve are excluded. Of the 21 included types, 8 (LUAD, LUSC, BRCA, COAD, BLCA, OV, PAAD, MESO) are represented in the cell model's training data and 13 were processed out of distribution; per-cancer tissue–cell discordance rates and the resulting caveats are reported in the main-text Methods.

Table S3: Definitions of the 38 histomic measurements. Machine-readable. Each row gives the measurement name, short and long descriptions, compartment, unit, computation strategy, and both category assignments (the six-way presentation grouping used in the main text and Figure 1, and the five-way grouping stored in the atlas metadata). The two schemes partition the identical 38 measurements and no measurement differs between them.

Presentation category (main text, Figure 1)	n
Tissue composition	3
Cell densities	8
Nuclear morphology and kinetics	7
Spatial organization	11
Spatial heterogeneity	5
Cell-type ratios	4
Atlas metadata category	**n**
Composition	3
Density	6
Morphology	8
Spatial	18
Heterogeneity	3

Full definitions: `histoatlas/config/feature_metadata.py`; served by the `list_features` tool and at https://histoatlas.com.

Table S4: Curated molecular target panel. Machine-readable. The 133 curated genes assessed for both expression and copy number, drawn from OncoKB and the COSMIC Cancer Gene Census, together with immune checkpoint targets and epithelial–mesenchymal transition and stemness markers; the 50 MSigDB Hallmark gene sets, all used in the per-cancer correlation analysis; and the 6 CIBERSORT immune cell fraction scores. File: `histoatlas/config/molecular_targets.py`.

Table S5: Benjamini–Hochberg correction family definitions. Machine-readable. One row per correction family, giving the family identifier, the analysis type, the grouping keys, the number of tests in the family, and the number significant at a false discovery rate below 0.05. Survival families are defined per cancer type × endpoint × model (149 families); correlation families per cancer type × molecular data type × model (150 families). Every adjusted P value in the atlas stores its family identifier and family size, so any reported result can be re-derived. Files: `correction_family_id` and `n_tests_in_family` columns of `survival_associations.parquet` and `feature_correlations.parquet`.

Table S6: Complete-case sample sizes and covariate missingness. Machine-readable. Per cancer type and endpoint: total patients, events, complete-case sample size, percentage dropped, per-covariate missing fractions for age, sex and pathologic stage, whether multiple imputation was used, the number of imputations, and the fraction of missing information. Files: `n_complete_case`, `pct_dropped`, `missing_frac_*`, `imputation_used`, `n_imputations` and `fmi` columns of `survival_associations.parquet`.

Table S7: Restricted mean survival time horizons and proportional-hazards violation prevalence. Machine-readable. Per cancer type: the restricted mean survival time horizon in days and its justification, and the proportion of fitted Cox models flagged pass, warn or fail on the Schoenfeld residual test. Across all 5,623 fitted models, 5,283 passed ($P \geq 0.05$), 270 warned ($0.01 \leq P < 0.05$) and 70 failed ($P < 0.01$); warned and failed models have their hazard ratios, confidence intervals and P values invalidated before correction. Files: `rmst_horizon_days` and `ph_flag` columns of `survival_associations.parquet`.

Table S8: Tests performed and significant results, by analysis type. Survival counts are Cox models; *valid* excludes models whose proportional-hazards assumption was violated. Significance is Benjamini–Hochberg adjusted $P < 0.05$ within the declared families.

Analysis	Stratum	Tests	Valid	Significant
Survival	Overall survival, unadjusted	759	723	60
Survival	Overall survival, adjusted	721	691	21
Survival	Progression-free survival, unadjusted	836	770	77
Survival	Progression-free survival, adjusted	760	705	23
Survival	Disease-specific survival, unadjusted	722	671	53
Survival	Disease-specific survival, adjusted	648	614	12
Survival	Disease-free survival, unadjusted	608	569	12
Survival	Disease-free survival, adjusted	569	540	2
Survival total		5,623	5,283	**260**
Correlation	Expression, unadjusted	117,436	117,436	28,182
Correlation	Expression, adjusted	85,649	85,649	22,466
Correlation	Pathway activity, unadjusted	42,150	42,150	12,397
Correlation	Pathway activity, adjusted	30,200	30,200	9,591
Correlation	Copy number, unadjusted	117,439	117,439	6,682
Correlation	Copy number, adjusted	85,652	85,652	6,027
Correlation	Immune scores, unadjusted	5,253	5,253	1,983
Correlation	Immune scores, adjusted	3,859	3,859	1,592
Correlation total		487,638	487,638	88,920

Table S9: CPTAC validation cohort and measurement concordance. A measurement passes when the standardized mean difference between cohort medians, scaled by the TCGA standard deviation, is at most 1.0. A cancer type passes when more than 80% of the 38 measurements pass ($\geq$ 31/38). Overall inter-measurement correlation geometry across the pooled cohorts: Pearson $r = 0.836$ on 703 upper-triangle pairs, $P = 6.1 \times 10^{-185}$ (prespecified criterion $r > 0.80$; passed).

TCGA	CPTAC	TCGA n	CPTAC slides	CPTAC patients	Within 1 SD	Pass
LUAD	LUAD	511	240	240	34/38 (89.5%)	yes
PAAD	PAAD	146	175	175	34/38 (89.5%)	yes
HNSC	HNSCC	471	390	112	32/38 (84.2%)	yes
BRCA	BRCA	1,037	198	198	31/38 (81.6%)	yes
UCEC	UCEC	459	92	92	30/38 (78.9%)	no
Total			1,095	**817**	**4/5 pass**	

Median tumor nuclear irregularity fails in all five pairs (standardized mean difference 1.65 to 2.34, largest in lung adenocarcinoma) and is flagged in the atlas as protocol-sensitive. The endometrial carcinoma failure is driven by tissue-composition shifts in the smallest cohort ($n = 92$ versus $n = 459$ in TCGA): stroma area fraction (1.44 SD lower), tumor area fraction (1.25 SD higher), stromal fibroblast density (1.29 SD lower) and tumor–stroma contact fraction (1.09 SD lower).

Table S10: Histomic program associations with protein abundance in CPTAC. Machine-readable, 2,220 program–protein pairs across four cancer types with proteomic profiles; 12.6% significant at $P < 0.05$. Strongest associations per program are given below. File: `data/paper/histomic_programs/program_protein_correlations.parquet`.

Program	Interpretation	Top protein correlates (Spearman ρ)
P1	Immune infiltration	CXCL9 (0.85), CD79A (0.82), CD3E (0.78)
P2	Stromal architecture	ZEB1 (0.72), FN1 (0.71)
P3	Nuclear atypia and proliferation	FGFR3 (-0.70), BRCA1 (-0.64)
P4	Tumor density	—
P5	Myeloid inflammation	—
P6	Fibroblast biology	—

Table S11: Benjamini–Hochberg versus Benjamini–Yekutieli correction. The Benjamini–Yekutieli procedure controls the false discovery rate under arbitrary dependence among tests and is therefore more conservative. For this comparison all tests of a given analysis type were pooled into a single correction family, which is why the survival denominator (272) differs from the 260 obtained under the production 149-family design reported in the main text.

Analysis	Benjamini–Hochberg	Benjamini–Yekutieli	Retention
Molecular correlations	88,920	60,576	68.1%
Survival associations	272	147	54.0%

Table S12: Sensitivity of leading correlations to molecular subtype adjustment. For each cancer type, the top 20 measurement–gene correlations were recomputed with molecular subtype as an additional covariate. Attenuation is the relative reduction in the median absolute Spearman coefficient.

Cancer type	Subtype variable	Pairs	Median $\|\rho\|$	Adjusted	Attenuation
BRCA	PAM50	20	0.584	0.558	4.3%
UCEC	Molecular class	20	0.406	0.381	6.2%
PAAD	Immune subtype (C1–C6)	20	0.375	0.269	28.1%
LUAD	*EGFR*/*KRAS* driver	20	0.579	—	not evaluable

The lung adenocarcinoma stratification was uninformative because neither *EGFR* nor *KRAS* mutation was prevalent enough in this cohort to form contrasting groups, leaving a single category. Pancreatic adenocarcinoma shows substantially stronger subtype confounding than breast or endometrial carcinoma; correlations in that cancer type should be interpreted with subtype adjustment.

Table S13: Sensitivity of survival associations to leukocyte-fraction adjustment. The 10 leading survival measurements were refitted with leukocyte fraction as an additional covariate, to test whether their prognostic signal is a surrogate for immune content. It is not: slightly more associations reach significance after adjustment. Machine-readable per-model results: `data/paper/sensitivity/purity_adjusted_survival.parquet` (358 fitted models).

Model	Significant ($P < 0.05$)	Retention
Base (clinical covariates)	25/179	—
Leukocyte-fraction adjusted	28/179	112%